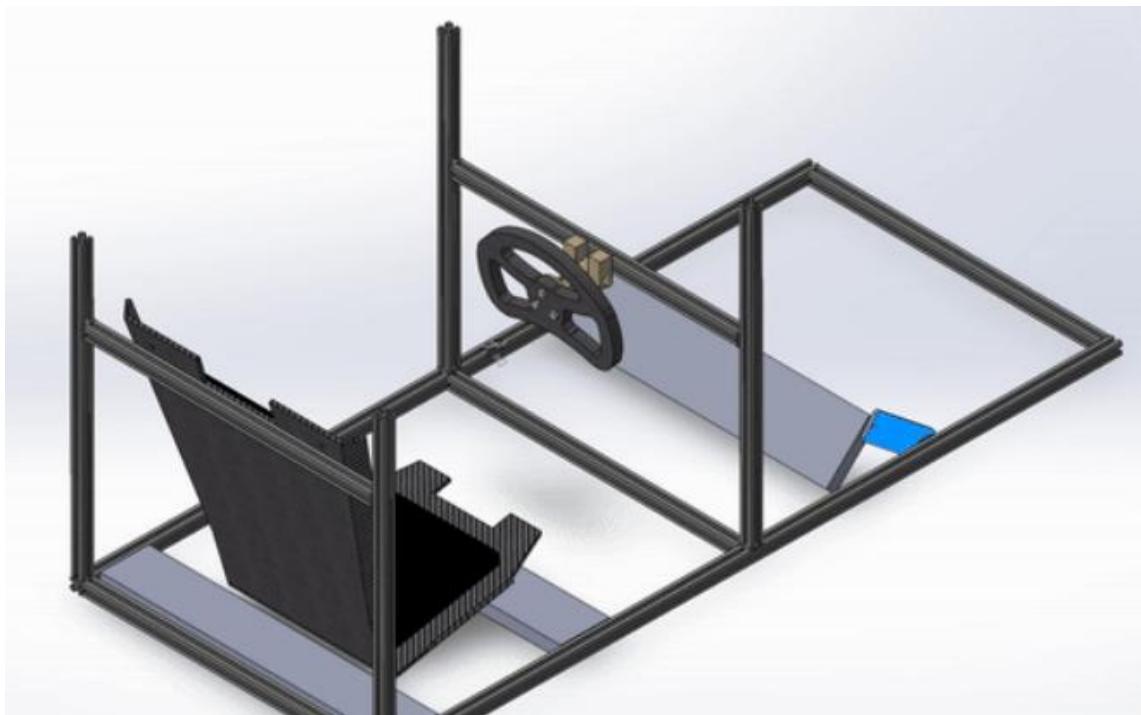


Ergonomics Jig

Validation for Clearance Models

Version 1.0 - July 20, 2026



Contents

Section 1 Project Overview	4
1.1 Background	4
1.2 Project Objective	5
1.3 My Role	5
1.4 Final Outcome	5
Section 2 Problem Definition	6
2.1 Existing Driver-Packaging Methods	6
2.2 Project Scope	Error! Bookmark not defined.
Section 3 Workflow Overview	7
3.1 Summarized Process	7
3.2 Software	8
3.3 LiDAR Background	8
3.4 Taking LiDAR Scans	Error! Bookmark not defined.
3.5 Manipulating LiDAR Scans	Error! Bookmark not defined.
3.5 Initial Demonstration	Error! Bookmark not defined.
3.6 Mesh Simplification	Error! Bookmark not defined.
3.7 Body Segmentation	Error! Bookmark not defined.
3.8 Model Assembly	Error! Bookmark not defined.
3.9 Clearance Shells	Error! Bookmark not defined.
Section 5 Collaboration and Integration	9
Section 6 Verification and Validation	11
Section 7 Results	13
Section 8 Workflow Iterations	14
4.1 The Loft Intersection	Error! Bookmark not defined.
4.2 Mesh-to-Part	Error! Bookmark not defined.
4.3 Blender Boolean	Error! Bookmark not defined.
Section 9 Future Improvements	14
9.1 Standardize LiDAR Scans	Error! Bookmark not defined.
9.2 Elaborate and Deliberate Mesh Processing	Error! Bookmark not defined.
9.3 Accurate Joint Placement	Error! Bookmark not defined.

9.4 Improved Clearance Geometry.....	Error! Bookmark not defined.
Section 10 References.....	14
Section 20 Document Revision History.....	15

Section 1 Project Overview

1.1 Background

“Driver packaging is essential to creating a competitive and rules-legal vehicle. Optimization in this area allows for minimum vehicle dimension, contributing to a lightweight vehicle.

Technical inspection for the competition, additionally, must be completed and passed in order to compete. In M24 (May 2025), only one driver was suitable for competition, as the rules state that laterally, “The driver’s helmet shall have 152 mm (6 in.) clearance, while the driver’s shoulders, torso, hips, thighs, knees, arms, elbows, and hands shall have 76 mm (3 in.) clearance.” Vertically, “The driver’s helmet shall have 152 mm (6 in.) minimum clearance from any two points among those members that make up to top of the roll cage” (Baja SAE Rules 2026).

M24 attempted to compete with multiple drivers for endurance (3hr race), and only one 5’5” driver was able to be accommodated, while others failed technical inspection.”

– From 1.1 of Clearance Models Documentation

In my Clearance Models documentation, I explain the development and implementation of LiDAR scans of potential drivers. To validate these models, and thus the chassis and subsystems developed using the models, an ergonomic jig was necessary.

The ergonomic jig allows for the ergonomic validation of the drivers, meaning the pedal and steering wheel placement can be verified to be within a comfortable, rules compliant, and integratable position.

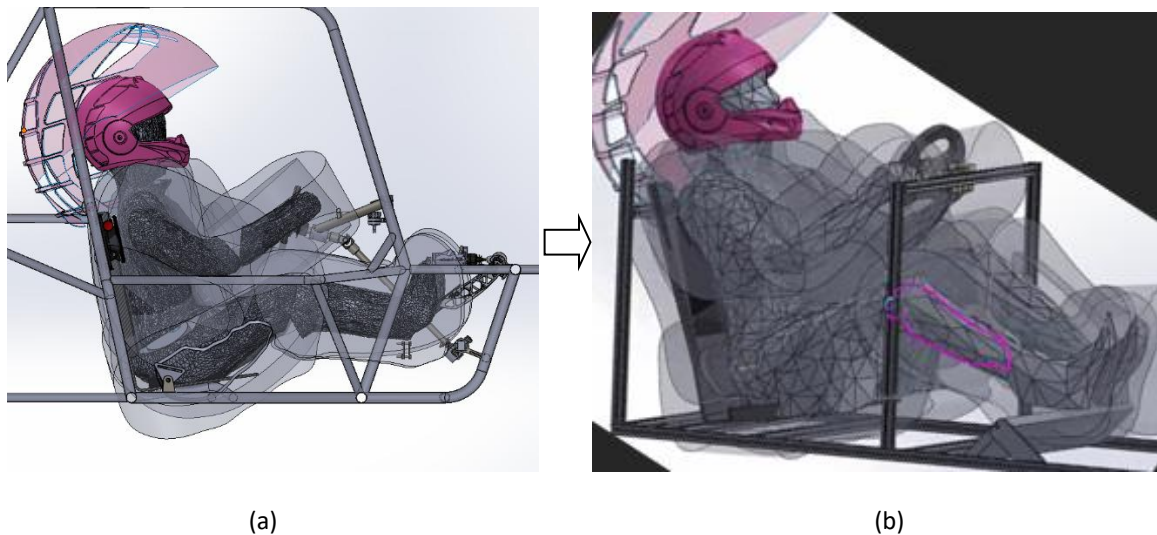


Figure 1: Clearance Model in (a) chassis and (b) ergonomic jig, demonstrating ergonomic jig capabilities with few parts while incorporating essential driver interface components (steering wheel, seat, and pedals).

1.2 Project Objective

I was tasked with validating the current CAD setup by developing and implementing an ergonomic jig. The jig is necessary to ensure model accuracy as well as ensure driver comfort, as despite the clearance models having limb angles within the ergonomic range set by NASA, ensuring driver comfort for the 3 hour race was essential.

Additionally, due to the scaling process associated with the clearance model generation, it was essential to determine accuracy of the models when compared to the real-life drivers.

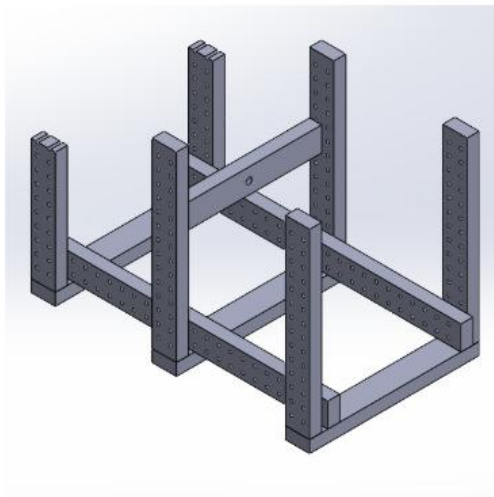


Figure 2: Previous year's (M24) ergonomic jig. Was not easily adjustable, moved in discrete intervals, and did not last a year when stored outdoors.

1.3 My Role

I was tasked with developing the mechanisms for ease of adjustability of the steering wheel and pedal. Allowing for quick and easy adjustments allows for greater accuracy through a more reliable mechanism. Determining the position for maximum comfort, maximum clearance, and the current configuration in CAD was required to be a quick and easy process, as time alters driver perception. If the adjustments take a significant amount of time, then the drivers may forget sensation, making their input unreliable.

1.4 Final Outcome

After creating and implementing the ergonomic jig, the quick process allowed for reliable driver input, as well as validation of the optimized setup in CAD, which also validates the accuracy of the clearance models, as the slight discrepancies of model to reality were deemed to be insignificant for our purposes.

Section 2 Engineering Requirements

2.1 Functional Requirements

The jig is required to validate the car's current chassis configuration, while also ensuring optimal steering wheel and pedal placement. It was therefore imperative that the steering wheel and pedal were easily adjustable to allow for ease of driver sensation comparison. The measurement procedure was also required to be quick and simple for the process to be viable.

2.2 Design Constraints

The cost of the jig had to remain as cheap as possible. However, the objective of the jig was to last for many years, allowing for yearly implementation so more engineering focus could be spent elsewhere in the future.

The system must be easily and continuously adjustable, measurable, and manufacturable.

2.3 Success Criteria

For the jig itself:

- The adjustments must be able to be performed under 15 seconds for each component, reliably.
- The measurements must be easily transferrable to CAD and have a reliable, repeatable setup.
- The frame of the jig must be sturdy enough for a driver to fall on a portion of it
 - Drivers' maximum weight was 150 lbs, assumed to fall from a height of 5'7" relative to the ground
- Adjustments must be continuous and stopable
- Jig must be able to last for 5+ years outdoors
- Driver swap must take less than 1 minute

Section 3 System Overview

3.1 Jig Architecture

The entire frame consists of 8020 Aluminum extrusions, which allows for T-slots with set screws to stop the bars in place. The static bars were fitted with 3D printed brackets. The steering wheel lies on an extrusion, which the bar itself is connected to two extrusions, allowing for lateral and vertical motion. The supporting extrusions are connected to the base, which also allows for forward and backward motion. The pedal, in this design, is connected to both base extrusions, as well as a link, allowing for translation and rotation. The seat, is mounted to an extrusion, allowing for different seat designs to be centered properly, and for angle adjustment through vertical translation of the extrusion.

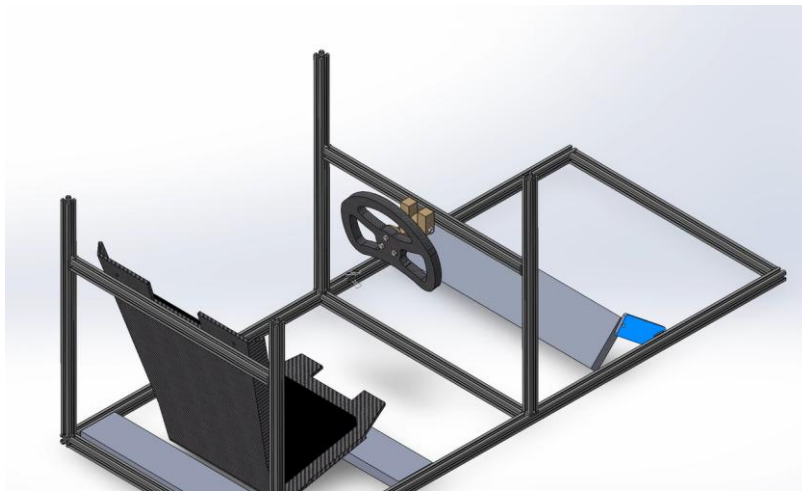


Figure 3: Ergonomic jig model demonstrating translation of the pedal measurement component.

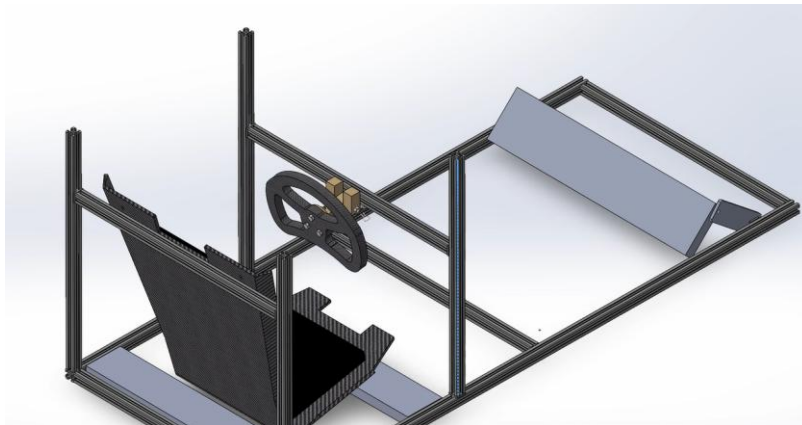


Figure 4: Ergonomic jig model demonstrating translation of the steering wheel measurement component, with pedal measurement component further back when compared to Figure 3.

3.2 Major Components

- 8020 Aluminum Extrusions
 - Used for jig frame and component translation and rotation
 - Significantly more durable and long-lasting than wood
- T-bolts
 - Fitted within the extrusion to allow for translation among the slotted portion of the extrusion. Tightening ceases movement capability
- 3D Printed Brackets
 - Used at static extrusion connecting points
- Steering Wheel Measurement Component
 - Used to measure potential steering wheel position based on comfort and chassis optimization
- Pedal Measurement Component
 - Used to measure potential pedal position based on comfort and chassis optimization

3.3 Bill of Materials (External)

Within this section is the BOM for only external components, meaning components requiring purchase, as the seat, steering wheel, and pedal were on hand, or manufacturable in the engineering makerspace for no cost.

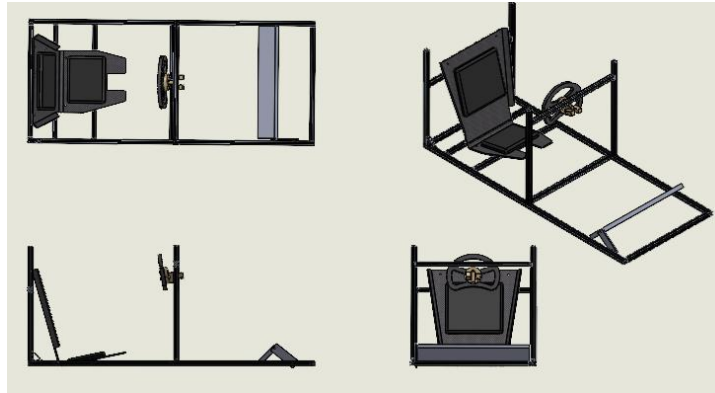


Figure 5: Top, Front, Right, and Isometric View of jig model.

Whole Dimensions for 8020 Rail Order		Measurement (Inches)	Measurement (Feet)	Each Cost (\$)	Quantity	Total Cost (\$)
Length		48	4	27.03	2	54.06
Width		24	2	11.67	4	46.68
Height of Seat Column		24	2	11.67	2	23.34
Height of Steering Column		24	2	11.67	2	23.34
					Total	147.42
Additional				Each Cost (\$)	Quantity	Total Cost (\$)
Perpendicular Pivot with T Slide				6.72	6	40.32
					Total	187.74

Figure 6: Total cost of extrusions and T-slots. Requirement was under \$250.

3.4 Measurement Procedure and Implementation

The measurement procedure is required to be quick and accurate, so clear points are necessary in the jig and in CAD. The distances must be easily measurable, as well as the angles.

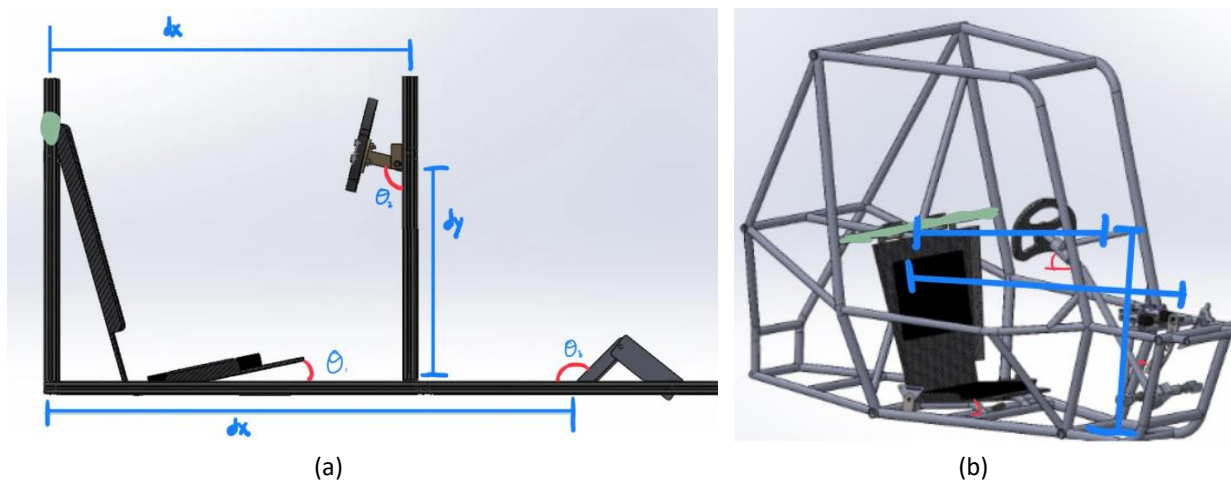


Figure 7: Proposed measurement system, presented for IDR. (a) Detailing of the measurements with respect to the top, back of the seat, relative to the components being measured horizontally. Y components measured relative to the basepoint of the seat, and angle measured from the horizontal or vertical. (b) Direct implementation to CAD.

Section 4 Steering Wheel Mechanism

4.1 Design Requirements

- Continuously adjustable laterally, forwards and backwards, and for the angle
- Easy to manufacture
- Easy to assemble
- Easy to stop at angle and position

4.2 Design Logic

The solution for the transverse motion was obvious, as the aluminum extrusions provided that capability. To actually mount the steering wheel to the extrusion, I initially thought to bolt the T-slots directly into the wheel, but that did not allow for rotational motion, so instead I focused on the rotational mechanism, as I could likely develop the mechanism and figure out a simple mounting method.

I wanted the design to be simple, so I thought of a set screw and its utilization of friction for stopping motion. I decided to have a plate which would be easy to mount the steering wheel onto, with a rectangular extrusion that had a hole in the center of it. Concentric to the hole would be two other holes with two other pieces that would directly mount to the aluminum extrusion. The two pieces mounted to the extrusion would have a fixed pose, while being able to translate left and right, with a relatively constant distance between them. The bolt connecting the three pieces would be able to be tightened, which would compress the center piece, stopping rotation.

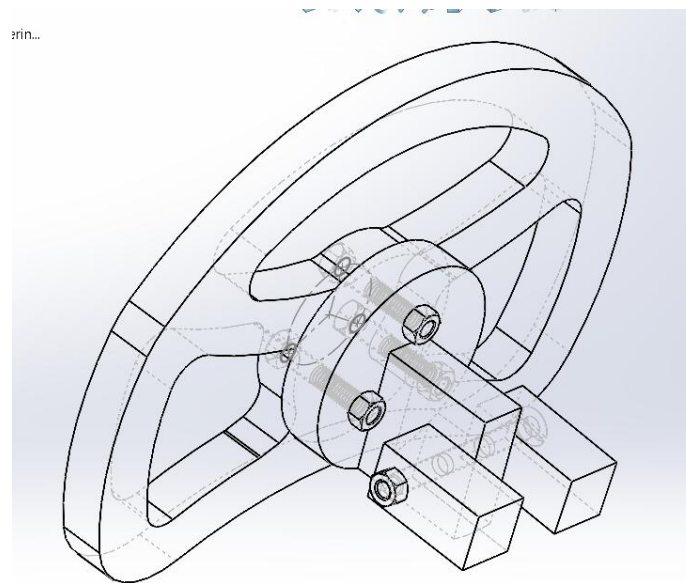


Figure 8: Steering wheel mechanism, showing attachment method and stopping mechanism.

4.3 Final Mechanism

The final mechanism worked well with practically no hiccups in any part of the process.

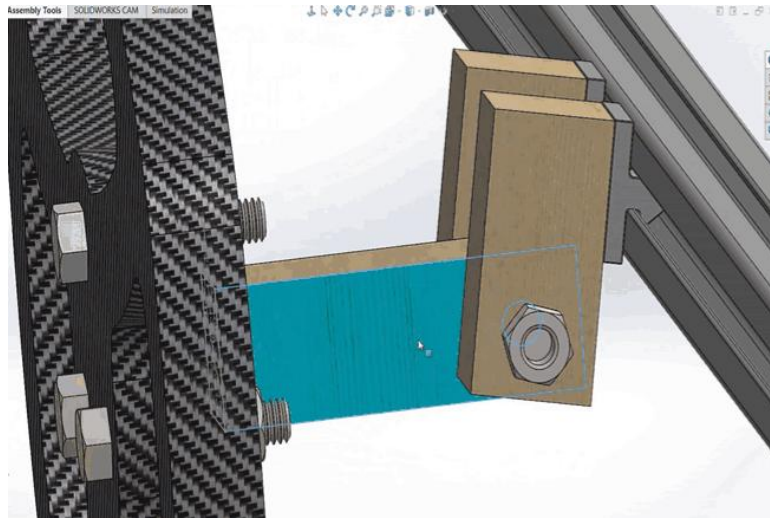


Figure 9: Close up of rotation mechanism, with T-slot attachments to compression blocks.

4.4 Manufacturing

Manufacturing was extremely straightforward. In the model, wood was chosen, as I believed we had a lot of spare, junk wood that could be utilized, but 3D printing was significantly easier. It was also free, as the makerspace is free to use. Filament itself likely only cost around \$2.

Assembly was also extremely easy, as proper clearance holes were created. Figure 9 does not show through holes for the compression blocks for the T-slots, but they were added in, and the blocks were decreased in thickness. The only difficult part of assembly was finding the correct bolts, as the state of the shop at that time was unsatisfactory to say the least.

Section 5 Pedal Mechanism

5.1 Design Requirements

- Continuously adjustable forwards and backwards, and for the angle
- Easy to manufacture
- Easy to assemble
- Easy to stop at an angle and position

5.2 Design Logic

The idea is to have the base of the system able to translate forwards and backwards. After, I thought of a method to rotate, and immediately I thought of hinges. A hinge attached to another part to rotate and translate seemed promising, but I couldn't figure out how to maintain the angle. A possible solution would be to put something underneath the pedal platform, but the idea was too "janky," and I wanted something significantly more elegant. I then thought of the mechanical door closers that alter the closing speed and time. I figured that I could add a link which would be rigid, allowing for better angle control. Stopping the link would also maintain the angle, so a simple component I added were binder clips, as they were available on hand and easy to implement. So, if a desired position and angle is achieved, then binder clips can stop the translation of both the base and the angle link.

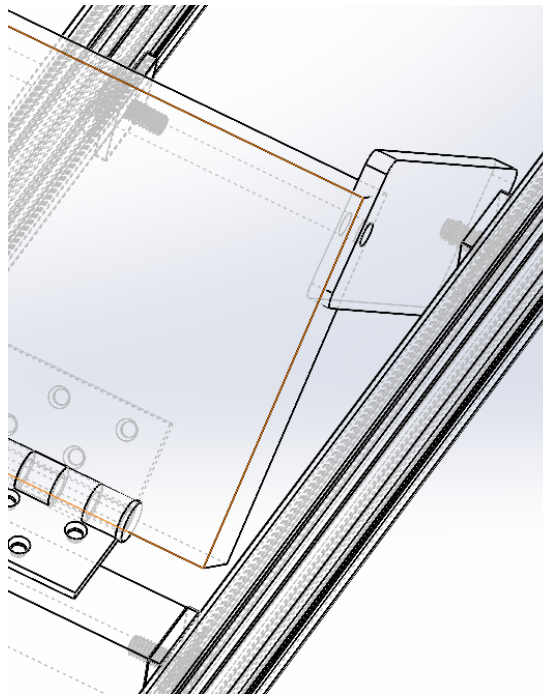


Figure 10: Pedal mechanism, showing the base, hinge attachment, platform, and rotation link. The binder clips are not shown in CAD.

5.3 Final Mechanism

The final mechanism worked well, though it was a bit trickier to line up the correct angle, as the driver had to input force, which slightly altered the static angle, increasing the time needed to fix and alter the setup.

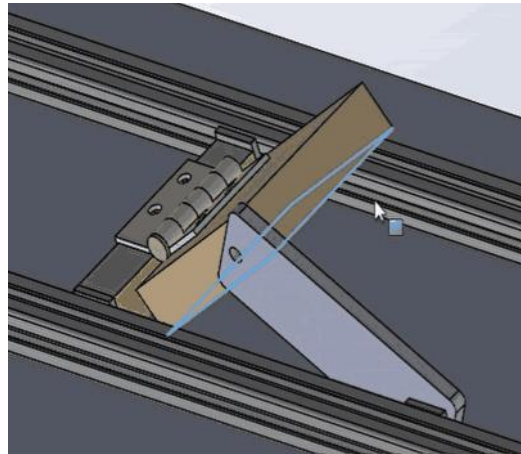


Figure 11: Pedal Mechanisms, final design.

5.4 Manufacturing

There is nothing significant to note about this process. Assembly was extremely straightforward. The platform and base were both made of wood, as I found them in the engineering dumpster. The hinge I bought from Home Depot, and the rotation link was 3D printed and rotated about a screw in the platform.

Section 6 Driver Validation Procedure

6.1 Validation Procedure

Drivers entered the ergonomic jig, which was preset to the CAD conditions. The drivers were then able to adjust the steering wheel and pedal position and angle till they determined a range of comfort and a range of optimal comfort.



Figure 12: Unfortunately we didn't take pictures of our jig for some reason, but here is how the driver would test the ergonomics. This model is not continuous, but ours is.

6.2 CAD Integration

Comparing the measurements to CAD made it clear that the models were accurate to half an inch, and that the current chassis design and subsystem integration was adequate. Optimizing for comfort, we took the data for the most comfortable ranges and found the overlap between the drivers. The drivers were then put into CAD to determine if the angle change in their limbs caused any clearance issues, or interference with their own body.

This information allowed me to determine steering wheel angle and pedal placement, though the steering wheel angle and position was more difficult, as I was later told I would have to minimize the U-joint angle, which is a joint attached to the steering wheel. This required me to compromise on some drivers' optimal comfort, as the only possible configuration necessitate a couple drivers to be in an ergonomic, but not optimal position. The decision of who the drivers would be was based on the probability that they would drive the complex courses, as we had a couple dedicated drivers for endurance (the 3 hour race).

Section 7 Results

The parts I contributed worked well and met all of the success criteria. The ergonomic jig as a whole functioned and provided us data, which we used to optimize driver comfort, while balancing the demands of other subteams and their constraints.

Section 8 Future Improvements

For our purposes, this ergonomic jig functioned well. However, it can always be better. For an easy implementation, it would be beneficial to add an easy measurement system, such as attached numbers for vertical and horizontal measurement. Protractors could also be attached for easy readings of angle.

To mirror the real version of the steering wheel, it would be beneficial to be able to turn the wheel. The implementation for this aspect is extremely simple and easy, but it was thought about after data was collected. It is unlikely that this feature would have impacted data significantly, however.

Also, the pedals on the car are two components, which also don't require the full foot to press on it. For more accurate angles and feel, attaching two pedals, which are spring loaded could be worthwhile.

Section 9 References

1. Baja Rules 2026 – Only PDF available (www.bajasae.net)
2. NASA -- Ergonomic Models of Anthropometry, Human Biomechanics, and Operator-Equipment Interfaces (<https://ntrs.nasa.gov/api/citations/19890017973/downloads/19890017973.pdf>)

Section 10 Document Revision History

Version	Date	Author	Change
1.0	7/20/2026	Nathan Robinson	Document Release